

Calculating absolute effects based on baseline risk and relative effects

Here we show three examples of how one can calculate a risk difference and its associated confidence interval given a relative risk (example 1), an odds ratio (example 2), or a hazard ratio (example 3) and a control group risk. Article 1 in this series provides a glossary explaining the technical terms in this Appendix including absolute risk, relative risk, hazard ratio, risk difference.

Example 1

The meta-analyzed relative risk (RR) of 'disability worsening' in patients with multiple sclerosis from two RCTs (n = 1,479) comparing teriflunomide to placebo proved to be 0.76 (95% CI 0.62 – 0.93).¹

The risk in the control group (estimated as the median risk in placebo groups) is 21%. When the number of included studies (denoted here as M) is even, the following formula can be used to calculate the median risk:

$$\frac{\text{control group event rate in } \frac{M}{2} \text{ th study} + \text{control group event rate in } (\frac{M}{2} + 1) \text{ th study}}{2}$$

Where the studies, M, are sorted by increasing order of event rate.

In case of two control groups (M=2), the median risk is just the average of the first (here, 18.4%) and second risk (here, 23.5%) in the control groups. The median Risk = (18.4% + 23.5%)/2 = 21%.

(1) Risk with intervention (*corresponding risk*) = RR x Risk with control
= 0.76 x 21%
= 16%

(2) Risk difference = Risk with control – Risk with intervention
= 21% – 16%
= 5%

We can use the same procedure to calculate the confidence interval (CI) associated with the risk difference, substituting the extremes of the CI (here, 0.62 and 0.93) for the point estimate (here, 0.76).

(3) For the upper limit of the RR CI (0.93):

- Risk with intervention = 0.93 x 21% = 19.5%
- Risk difference = Risk with control – Risk with intervention = 21% – 19.5% = 1.5%

(4) For the lower limit of the RR CI (0.62):

- Risk with intervention = 0.62 x 21% = 13.0%
- Risk difference = Risk with control – Risk with intervention = 21% – 13.0% = 8.0%

(5) Risk difference is: 5% (95% CI: 1.5% to 8.0%).

Example 2

The odds ratio (OR) of 'incident hypertension' in pregnant women with overweight or obesity from four RCTs (n = 1,324) comparing exercise therapy + usual care to usual care is 0.52 (95% CI 0.28 – 0.96).²

The risk in the control group (estimated as the median risk in usual care groups) is 14%. When the number of included studies is even, the formula to calculate the median risk mentioned in example 1 can be applied. In case of four control groups (M=4), the median risk is just the average of the second (here, 9%) and third risk (here, 19.1%) in the control groups. The median risk = (9% + 19.1%)/2 = 14%.

To convert an odds ratio to a RR the next formula³ can be used:

$$\begin{aligned} (1) \quad RR &= \frac{OR}{(1 - \text{risk in control group}) + (\text{risk in control group} \times OR)} \\ &= \frac{0.52}{(1 - 0.14) + (0.14 \times 0.52)} \\ &= 0.56 \end{aligned}$$

We can use the same procedure to calculate to confidence interval (CI) associated with the odds ratio, substituting the extremes of the CI (here, 0.28 and 0.96) for the point estimate (here, 0.52).

(2) For the upper limit of the OR CI (0.96):

$$\bullet \quad RR = \frac{0.96}{(1 - 0.14) + (0.14 \times 0.96)} = 0.97$$

(3) For the lower limit of the OR CI (0.28):

$$\bullet \quad RR = \frac{0.28}{(1 - 0.14) + (0.14 \times 0.28)} = 0.31$$

Now we have the RRs of the point estimate (0.56) and its associated confidence interval (0.31 – 0.97) for a risk in the control group (estimated as the median risk in placebo groups) of 14%.

(4) Risk with intervention (*corresponding risk*) = RR x Risk with control

$$\begin{aligned} &= 0.56 \times 14\% \\ &= 7.8\% \end{aligned}$$

(5) Risk difference = Risk with control – Risk with intervention

$$\begin{aligned} &= 14\% - 7.8\% \\ &= 6.2\% \end{aligned}$$

We can use the same procedure to calculate the confidence interval (CI) associated with the risk difference, substituting the extremes of the CI (here, 0.31 and 0.97) for the point estimate (here, 0.56).

(6) For the upper limit of the RR CI (0.97):

- Risk with intervention = 0.97 x 14% = 13.6%
- Risk difference = Risk with control – Risk with intervention = 14% – 13.6% = 0.4%

(7) For the lower limit of the RR CI (0.31):

- Risk with intervention = 0.31 x 14% = 4.3%
- Risk difference = Risk with control – Risk with intervention = 14% – 4.3% = 9.7%

Risk difference is: 6.2% (95% CI: 0.4% to 9.7%).

Example 3

The hazard ratio (HR) of survival in patients with advanced hepatocellular carcinoma (n = 602) comparing sorafenib to placebo is 0.69 (95% CI 0.55 – 0.87).⁴ Patients were followed up to about 17 months. We present two calculations. The first calculation relates to *event-free survival*. The second calculation relates to an *event*, i.e. death in this example.

The proportion of event-free patients up to about *eight* months in the control group is 0.50 (50%). To convert a hazard ratio (HR) to a proportion of event-free patients in the intervention group the next formula⁵ can be used:

$$\begin{aligned} \text{(1) Proportion with intervention} &= e^{\ln(\text{control group event-free proportion}) \times \text{HR}} \\ &= e^{\ln(0.50) \times 0.69} \\ &= e^{-0.6931 \times 0.69} \\ &= e^{(-0.4783)} \\ &= 0.6199 \text{ (i.e. 62.0\% when rounded)} \end{aligned}$$

$$\begin{aligned} \text{(2) Event-free survival difference} &= \text{proportion event-free patients with intervention} - \text{proportion event-free patients with control} \\ &= 62\% - 50\% = 12.0\% \text{ up to about eight months} \end{aligned}$$

We can use the same procedure to calculate the confidence interval (CI) associated with the hazard ratio, substituting the extremes of the CI (here, 0.55 and 0.87) for the point estimate (here, 0.69).

$$\begin{aligned} \text{(3) For the upper limit of the CI (0.87):} \\ \bullet \text{ Proportion with intervention} &= e^{\ln(0.50) \times 0.87} = 0.5471 \text{ (i.e. 54.7\% rounded)} \\ \bullet \text{ Event-free survival difference} &= \text{proportion event-free patients with intervention} - \text{proportion event-free patients with control} \\ &= 54.7\% - 50\% = 4.7\% \end{aligned}$$

$$\begin{aligned} \text{(4) For the lower limit of the CI (0.55):} \\ \bullet \text{ Proportion with intervention} &= e^{\ln(0.50) \times 0.55} = 0.6830 \text{ (i.e. 68.3\% rounded)} \\ \bullet \text{ Event-free survival difference} &= \text{proportion event-free patients with intervention} - \text{proportion event-free patients with control} \\ &= 68.3\% - 50\% = 18.3\% \end{aligned}$$

$$\text{(5) Event-free survival difference up to about eight months is: 12.0\% (95\% CI: 4.7\% to 18.3\%).}$$

The next section addresses calculating the occurrence of an event, i.e. death in this scenario.

As mentioned earlier, the second calculation relates to the occurrence of an event, i.e. death in this example; we use the same data as used for event-free survival. The proportion of patients with event in the control group [i.e. control group risk] up to about *ten* months is 0.625 (62.5%).

To convert a hazard ratio (HR) to an absolute risk in the intervention group the next formula⁵ can be used:

$$\begin{aligned} \text{(1) Risk with intervention} &= 1 - e^{\ln(1 - \text{control group proportion events}) \times HR} \\ &= 1 - e^{\ln(1 - 0.625) \times 0.69} \\ &= 1 - 0.5083 = 0.4917 \text{ (i.e. 49.2\% when rounded)} \end{aligned}$$

$$\begin{aligned} \text{(2) Risk difference} &= \text{Risk with control} - \text{Risk with intervention} \\ &= 62.5\% - 49.2\% \\ &= 13.3\% \text{ up to about ten months} \end{aligned}$$

We can use the same procedure to calculate the confidence interval (CI) associated with the hazard ratio, substituting the extremes of the CI (here, 0.55 and 0.87) for the point estimate (here, 0.69).

(3) For the upper limit of the CI (0.87):

- Risk with intervention = $1 - e^{\ln(1 - 0.625) \times 0.87} = 0.5740$ (i.e. 57.4% rounded)
- Risk difference = Risk with control – Risk with intervention = 62.5% – 57.4% = 5.1%

(4) For the lower limit of the CI (0.55):

- Risk with intervention = $1 - e^{\ln(1 - 0.625) \times 0.55} = 0.4169$ (i.e. 41.7% rounded)
- Risk difference = Risk with control – Risk with intervention = 62.5% – 41.7% = 20.8%

(5) Risk difference up to about ten months is: 13.3% (95% CI: 5.1% to 20.8%).

References

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